Homework 4

EN.601.439/639 | Fall 2026

You must submit either a physical handwritten homework or print out your homework to submit at the beginning of class.

Problem 1: Equivalence of k-means objectives

Given data points $x_1, \ldots, x_N \in \mathbb{R}^D$ and a number K of clusters, the K-means problem is to compute K "centroids" $w_1, \ldots, w_K \in \mathbb{R}^D$ that minimize the sum of squared errors:

$$\arg\min_{w_1,\ldots,w_K \in \mathbb{R}^D} \sum_{i=1}^{N} \left(\min_{k=1,\ldots,K} ||x_i - w_k||^2 \right). \tag{1}$$

Consider the following optimization problem to compute centroid vectors $w_1, \ldots, w_K \in \mathbb{R}^D$ *and* binary indicators $z_{ik} \in \{0, 1\}$ for $i = 1, \ldots, N$ and $k = 1, \ldots, K$ that minimize the following objective:

$$\arg\min_{\substack{w_1,\ldots,w_K \in \mathbb{R}^D \\ z_{11},\ldots,z_{NK} \in \{0,1\} \\ \sum_{k=1}^{K} z_{ik} = 1 \text{for } i=1,\ldots,N}} \sum_{i=1}^{N} \sum_{k=1}^{K} ||x_i - w_k||^2 \cdot z_{ik}. \tag{2}$$

a) Suppose $(\widehat{w}_1, \ldots, \widehat{w}_K)$ is a solution to (1). Prove that there exist $\widehat{z}_{11}, \ldots, \widehat{z}_{NK}$ such that $(\widehat{w}_1, \ldots, \widehat{w}_K, \widehat{z}_{11}, \ldots, \widehat{z}_{NK})$ is a solution to (2).

b) Conversely, suppose that $(\widehat{w}_1, \ldots, \widehat{w}_K, \widehat{z}_{11}, \ldots, \widehat{z}_{NK})$ is a solution to (2). Then prove that $\widehat{w}_1, \ldots, \widehat{w}_K$ is a solution to (1).

Problem 2: EM algorithm for Gaussian mixture models

We are given one-dimensional data points $x_1, \ldots, x_N \in \mathbb{R}$, where each x_i is independently sampled from a Gaussian mixture model (GMM) with K components:

$$x_i \sim \pi_1 \cdot N(\mu_1, \sigma_1^2) + \pi_2 \cdot N(\mu_2, \sigma_2^2) + \cdots + \pi_K \cdot N(\mu_K, \sigma_K^2), \tag{3}$$

where $\pi_k > 0$ is the mixture proportion for the k-th component $N(\mu_k, \sigma_k^2)$ with mean $\mu_k \in \mathbb{R}$ and variance $\sigma_k^2 > 0$. (We require $\sum_{k=1}^K \pi_k = 1$.)

Equivalently, we may describe each data point x_i as being sampled from a hierarchical model where (1) we sample a latent variable $z_i \sim \text{Categorical}(\pi_1, \ldots, \pi_K)$ indicating the mixture component for data point i and (2) sample $x_i \sim N(\mu_{z_i}, \sigma_{z_i}^2)$.

We will derive the two steps of the EM algorithm to estimate the parameters $\Theta = (\pi_k, \mu_k, \sigma_k^2)_{k=1,\ldots,K}$ from the observed data $x_1, \ldots, x_N$.

a) In the E-step of the EM algorithm, given parameters Θ, we compute the probability $P(Z_i = k \mid X_i, \Theta)$ of each data point x_i being sampled from the k-th mixture component. Using Bayes' law, prove that

$$P(Z_i = k \mid X_i, \Theta) = \frac{\pi_k \cdot f(X_i; \mu_k, \sigma_k^2)}{\sum_{k'=1}^K \pi_{k'} \cdot f(X_i; \mu_{k'}, \sigma_{k'}^2)}. \tag{4}$$

We call the quantity $\gamma_{ik} \triangleq P(Z_i = k \mid X_i, \Theta)$ the *responsibility* of Gaussian k for data point i.

b) In the M-step of the EM algorithm, given responsibilities γ_{ik}, we aim to compute mixture proportions π_k, means μ_k, and variances σ_k^2 that maximize the following objective:

$$\underset{\substack{\pi_k > 0 \\ \sum_{k=1}^K \pi_k = 1 \\ \mu_k \in \mathbb{R}, \sigma_k^2 > 0}}{\text{argmax}} \sum_{i=1}^N \sum_{k=1}^K \gamma_{ik} \cdot \log\left(\pi_k f(x_i; \mu_k, \sigma_k^2)\right), \tag{5}$$

where $f(x_i; \mu_k, \sigma_k^2) = \frac{1}{\sqrt{2\pi\sigma_k^2}} \cdot e^{-\frac{(x-\mu_k)^2}{2\sigma_k^2}}$ is the PDF of the Gaussian distribution $N(\mu_k, \sigma_k^2)$.

Prove that the solution to (5) is given by:

$$\widehat{\mu}_k = \frac{\sum_{i=1}^N \gamma_{ik} x_i}{\sum_{i=1}^N \gamma_{ik}} \tag{6}$$

$$\widehat{\sigma^2}_k = \frac{\sum_{i=1}^N \gamma_{ik} \cdot (x_i - \widehat{\mu}_k)^2}{\sum_{i=1}^N \gamma_{ik}} \tag{7}$$

$$\widehat{\pi}_k = \frac{\sum_{i=1}^N \gamma_{ik}}{N}. \tag{8}$$

[Hint: For the first two equations, differentiate the objective of (5) with respect to μ_k or σ_k^2, set equal to 0, and solve. For the third equation, plug in $\pi_1 = 1 - \sum_{k=2}^K \pi_k$; then differentiate with respect to π_k for each $k = 2, \ldots, K$, set equal to 0, and solve.]

Problem 3: Hidden Markov random fields and the iterated conditional modes (ICM) algorithm

We are given data points $x_1, \ldots, x_N$, a number K of clusters. Let $G = (V, E)$ be a (undirected) graph on vertices $V = \{1, \ldots, N\}$ and let $\beta > 0$.

We assume the observed data points $x_1, \ldots, x_N \in \mathbb{R}^D$ are generated in the following way:

1. Cluster labels $\mathbf{z} = (z_1, \ldots, z_N) \in \{1, \ldots, K\}^N$ are sampled from a distribution (called a *Potts model*):
$$p(z_1, \ldots, z_N) = \frac{1}{Z} \exp\left(\beta \sum_{(i,j)\in E} 1(z_i = z_j)\right). \tag{9}$$
2. Each data point x_i, conditioned on its cluster label $z_i = k$, is independently drawn from a Gaussian distribution:
$$x_i \mid z_i = k \sim N(\mu_k, \Sigma_k) \tag{10}$$
for some mean vector $\mu_k \in \mathbb{R}^D$ and covariance matrix $\Sigma_k \in \mathbb{R}^{D\times D}$.

That is, there are K clusters defined by Gaussian distributions $N(\mu_1, \Sigma_1), \ldots, N(\mu_K, \Sigma_K)$, respectively, where each latent variable z_i indicates the cluster that data point i belongs to and β controls how likely it is that two data points adjacent in the graph G have the same label. Note that the data points x_i are conditionally independent given the latent variables $\mathbf{z}$.

a) First, suppose we are given the cluster means μ_k and covariances Σ_k. The maximum a posteriori (MAP) estimation problem for estimating the cluster labels $\mathbf{z}$ is given by
$$\operatorname*{argmax}_{\mathbf{z}\in\{1,\ldots,K\}^N} p(\mathbf{z} \mid x_1, \ldots, x_N). \tag{11}$$
Show that (11) is equivalent to the following optimization problem
$$\operatorname*{arg\,min}_{\mathbf{z}\in\{1,\ldots,K\}^N} \frac{1}{2}\sum_{i=1}^{N}\left(\sum_{k=1}^{K}\left((x_i-\mu_k)^\top\Sigma_k^{-1}(x_i-\mu_k) + \log\det\Sigma_k\right)\cdot 1(z_i=k)\right) - \beta\sum_{(i,j)\in E} 1(z_i = z_j). \tag{12}$$

b) How many possible values can the cluster labels $\mathbf{z}$ in (12) take? Explain why it is difficult to directly compute the cluster labels $\mathbf{z}$ that optimize (12).

c) Suppose we want to optimize a *single* cluster label z_i, while fixing all of the other labels $\mathbf{z}_{-i} = (z_1, \ldots, z_{i-1}, z_{i+1}, \ldots, z_N)$. Derive a procedure for updating the cluster label z_i.

The procedure of iteratively updating the cluster labels z_i is called the *iterated conditional modes (ICM)* algorithm.

d) Now suppose the cluster means μ_k and covariances Σ_k are *unknown*. Derive an alternating algorithm to solve for both the cluster labels $\mathbf{z}$ and cluster parameters μ_k, Σ_k. Your algorithm should:

- Fix the cluster parameters μ_k, Σ_k and use ICM to update the cluster labels $\mathbf{z}$.
- Fix the cluster labels $\mathbf{z}$ and update the cluster parameters μ_k, Σ_k. (What should the update rule be?)